COS 433: Cryptography

Announcements:

- Problem Set 4 due today

Last time:

1. Random oracle heuristic
2. RSA signatures
3. Beginning of Schnorr signatures

Schnorr/DSA Signatures

Signature scheme defined over:

- Group G of size p
  - In practice: “elliptic curve group” (ECDSA)
  - Given generator g for G
- Scheme is secure in the ROM, assuming the hardness of discrete log

Schnorr Signatures

$\text{sk} = x \leftarrow \mathbb{Z}_p$
$\text{vk} = g^x \in G$



$m \in \{0,1\}^*$



$y = g^x$

$(m, R = g^r, z = H(R,m) \cdot x + r)$



$\text{Verify}(y, m, R, z) = 1$ if $g^z = y^{H(R,m)} \cdot R$

How should we sign messages?

- Idea: we need to do something that requires knowledge of x <u>but does not reveal</u> x
- Actual scheme: uses a hash function H to determine how to mask x
  - Set $R = g^r$
  - Compute $s = H(R,m)$
  - Set $\sigma = (R, s \cdot x + r)$

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Why is this secure?

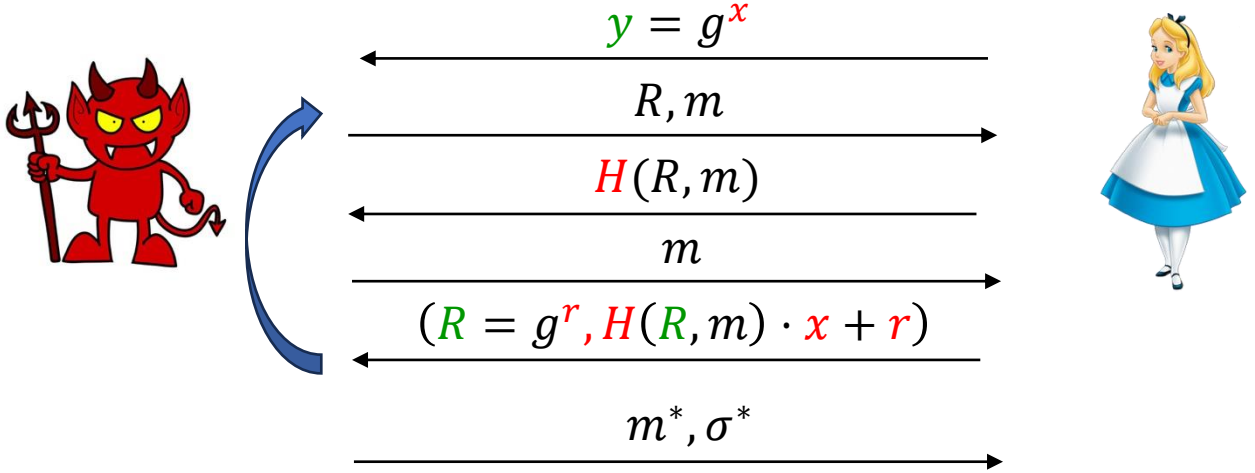
Schnorr Signatures

Key point #1: ”signing queries are useless to the adversary”

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Security (random oracle model): for all efficient adversaries $A^{H(\cdot),\mathrm{Sign}(\mathrm{sk},\cdot)}$



$$\Pr_{x,H}[m^* \text{ not queried to } \mathrm{Sign}(\mathrm{sk},\cdot) \text{ and forgery}] = \mathrm{negl}(\lambda)$$

Schnorr Signatures

Key point #1: ”signing queries are useless to the adversary”

Security (random oracle model): for all efficient adversaries $A^{H(\cdot),\mathrm{Sign}(\mathrm{sk},\cdot)}$



$$y = g^x$$

$$m$$

$$(R = g^r, z = H(R,m) \cdot x + r)$$



To answer a signing query:

- Sample r
- Generate $s = H(R,m)$
- Compute $z = s \cdot x + r$

Schnorr Signatures

Key point #1: ”signing queries are useless to the adversary”

Security (random oracle model): for all efficient adversaries $A^{H(\cdot),\mathrm{Sign}(\mathrm{sk},\cdot)}$



$$y = g^x$$

$$m$$

$$(R = g^r, z = H(R,m) \cdot x + r)$$



To answer a signing query (this looks identical):

- Sample uniform z
- Sample uniform s
- Set $R = g^z \cdot y^{-s}$
- Define $H(R,m) = s$

“Signatures are not telling you anything useful”

Schnorr Signatures



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$(m, R = g^r, z = H(R,m) \cdot x + r)$



$\mathrm{Verify}(y, m, R, z) = 1$ if $g^z = y^{H(R,m)} \cdot R$

0-time security (random oracle model): for all efficient adversaries $A^{H(\cdot)}$



y

R, m

$H(R,m)$

m^*, R^*, z^*



How to prove?

$$\Pr_{x,H}\left[g^{z^*} = y^{H(R^*,m^*)} \cdot R^*\right] = \mathrm{negl}(\lambda)$$

Schnorr Signatures

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$\mathrm{Verify}(y, m, R, z) = 1$ if $g^z = y^{H(R,m)} \cdot R$

Trivial attack:



y

m, R

$H(R,m)$

z



$$\Pr_{x,H}\left[g^z = y^{H(R,m)} \cdot R\right] = \mathrm{negl}(\lambda)$$

Schnorr Signatures

$\mathrm{sk} = x \leftarrow \mathbb{Z}_p$
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$\mathrm{Verify}(y, m, R, z) = 1$ if $g^z = y^{H(R,m)} \cdot R$

Trivial attack (equivalent):



y

m, R

s

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$$\Pr_{x,s}[g^z = y^s \cdot R] = \mathrm{negl}(\lambda)$$

Schnorr Signatures



$\mathrm{sk} = x \leftarrow \mathbb{Z}_p$

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Trivial attack (equivalent):



y

R

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z



$\Pr_{x,s}[g^z = y^s \cdot R] = \mathrm{negl}(\lambda)$

Observe: if $R = 1$ this is a discrete log attack!

Schnorr Signatures

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Analysis:

- Lemma: if the scheme is secure against trivial attacks, then it is secure in the ROM
  - Proof: run  guess which (R, m) query to H gives  the forgery
  - Answer the other queries ourselves (“backwards sampling”)
- Lemma: if discrete log is hard, this scheme has no trivial attacks
  - Proof: deferred

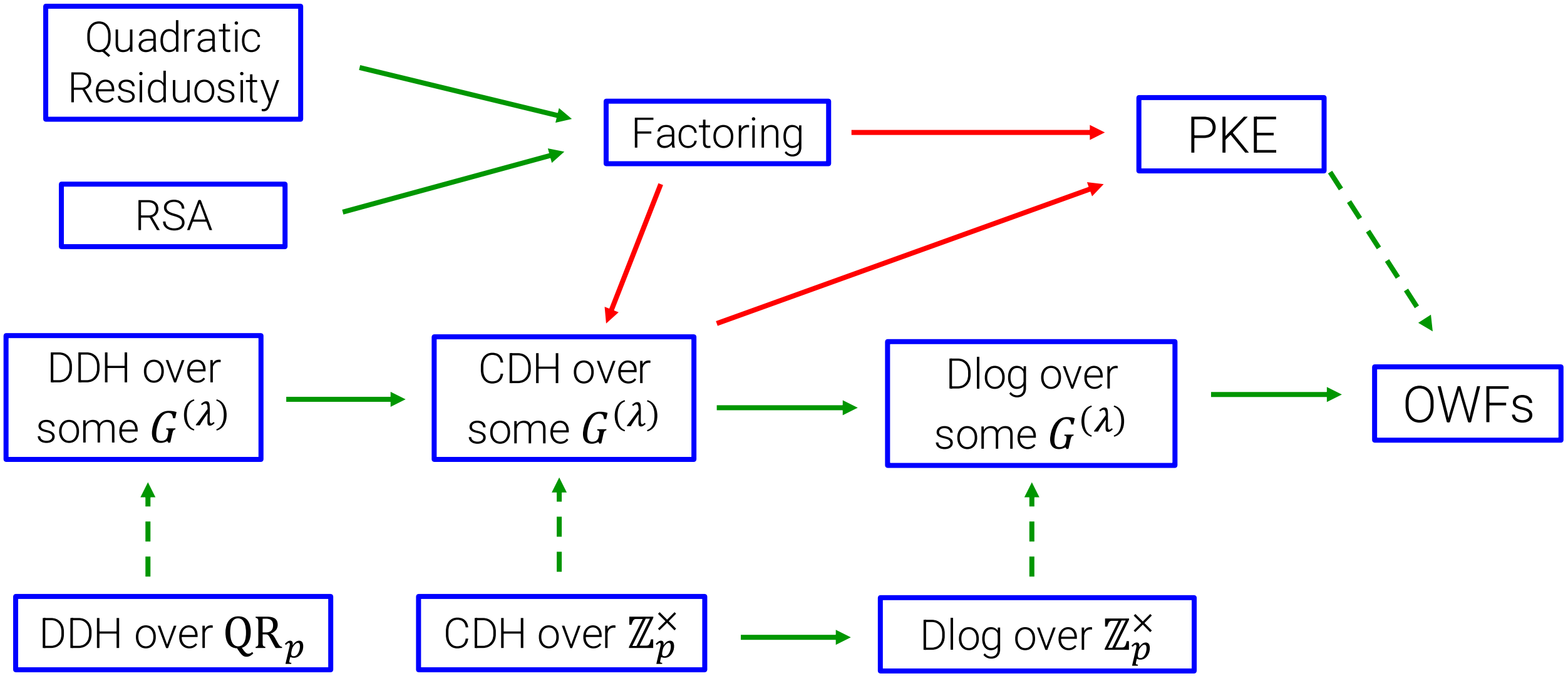
End of digital signatures!

1980s Public-Key Crypto Recap

- Public-key Encryption from number theory (RSA/QR/Factoring, Diffie-Hellman)

- Digital signatures:
  - 1-time security from CRHFs
  - Efficient many-time secure schemes from (RSA or discrete log) + random oracle

Number-Theoretic Hardness Assumptions



Part III of the course: “post-modern crypto”

Beyond Encryption + Authentication

What is cryptography for?

- Mathematically or computationally enforcing expected behavior
- Enabling trust even in the presence of untrustworthy parties

Encryption



k, m

$\mathrm{ct} = \mathrm{Enc}(k, m)$





k

$m' = \mathrm{Dec}(k, \mathrm{ct})$

- Secret communication in the presence of an untrusted eavesdropper

Authentication



k, m



$(m, \sigma = \text{Sign}(k, m))$



k

$\text{Verify}(k, m, \sigma) \in \{0, 1\}$

- Authentic communication in the presence of an (untrusted) impersonator

Beyond Encryption and Authentication





What if Alice and Bob do not trust each other?

What do they even want to accomplish?

Beyond Encryption and Authentication





Next 2-3 lectures: proof systems

Proofs

Ax. 1. $(P(\varphi) \wedge \Box\,\forall x(\varphi(x) \Rightarrow \psi(x))) \Rightarrow P(\psi)$

Ax. 2. $P(\neg\varphi) \Leftrightarrow \neg P(\varphi)$

Th. 1. $P(\varphi) \Rightarrow \Diamond\,\exists x\,\varphi(x)$

Df. 1. $G(x) \Leftrightarrow \forall\varphi(P(\varphi) \Rightarrow \varphi(x))$

Ax. 3. $P(G)$

Th. 2. $\Diamond\,\exists x\,G(x)$

Df. 2. $\varphi \text{ ess } x \Leftrightarrow \varphi(x) \wedge \forall\psi\,(\psi(x) \Rightarrow \Box\,\forall y(\varphi(y) \Rightarrow \psi(y)))$

Ax. 4. $P(\varphi) \Rightarrow \Box\,P(\varphi)$

Th. 3. $G(x) \Rightarrow G \text{ ess } x$

Df. 3. $E(x) \Leftrightarrow \forall\varphi(\varphi \text{ ess } x \Rightarrow \Box\,\exists y\,\varphi(y))$

Ax. 5. $P(E)$

Th. 4. $\Box\,\exists x\,G(x)$



Proofs

Ax. 1. $(P(\varphi) \wedge \Box\,\forall x(\varphi(x) \Rightarrow \psi(x))) \Rightarrow P(\psi)$

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Traditional/axiomatic proofs

A seminar??

Proofs



π



Prover

Verifier

CS insight: think of a proof as a form of computation

Proofs



Claim: $\forall x \forall y \forall z (\neq 0)$,
$\forall n > 2, x^n + y^n \neq z^n$

π



Prover

Verifier

CS insight: think of a proof as a form of computation

E.g.: AW writes a proof in a paper, Bob can then read the paper and check it.

Proofs



Claim: $\forall x \forall y \forall z (\neq 0)$,
$\forall n > 2, x^n + y^n \neq z^n$



π

Prover

Verifier



A seminar talk:

- Does not provide an entire, formal proof
- Goal (one of many): convince the audience that a formal proof exists
- Leverages audience interaction to make the proof easier to check